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Group: SE-1
lab2: Git Practice Exercises

I2-IP-group

https://github.com/pressamchara-art/lab1.git

Practice 1. Initialize and Commit a New Java Project

```
User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ git init
Reinitialized existing Git repository in D:/GitHello/.git/

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ nano GitHello.java

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ git add GitHello.java

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ git commit -m "Initial commit: GitHello programming"
[master 3b1a67c] Initial commit: GitHello programming
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Practice 2. Create and Switch Branches

```
User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ git checkout  feature/greeting
Switched to branch 'feature/greeting'

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (feature/greeting)
$ nano GitHello.java

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (feature/greeting)
$ git add GitHello.java

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (feature/greeting)
$ git commit -m "Add personalized greeting feature"
[feature/greeting 8a4ffb5] Add personalized greeting feature
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Practice 3. Merge a Branch

```
User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (feature/greeting)
$ git checkout master
Switched to branch 'master'

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ git merge  feature/greeting
Auto-merging GitHello.java
CONFLICT (content): Merge conflict in GitHello.java
Automatic merge failed; fix conflicts and then commit the result.

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master|MERGING)
$ git log --oneline
3b1a67c (HEAD -> master) Initial commit: GitHello programming
e4ed821 Welcome to our project
1a6dcde Add personalized greeting feature
59d9b9e Add personalized greeting feature
1b5c1f7 Initial commit: GitHello programing

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master|MERGING)
$ nano GitHello.java

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master|MERGING)
$ git add GitHello.java

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master|MERGING)
$ git commit -m "Welcome to our project"
[master 6891530] Welcome to our project

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ git log --oneline
6891530 (HEAD -> master) Welcome to our project
8a4ffb5 (feature/greeting) Add personalized greeting feature
3b1a67c Initial commit: GitHello programming
e4ed821 Welcome to our project
1a6dcde Add personalized greeting feature
59d9b9e Add personalized greeting feature
1b5c1f7 Initial commit: GitHello programing
```

Practice 4. Undo Mistakes with git reset and git revert

```
User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ git revert -m 1 HEAD
[master f789b78] Revert "Welcome to our project"
 1 file changed, 4 deletions(-)

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ git log
commit f789b7832d14ce1521ad6102873ebb38d3a4f2b2 (HEAD -> master)
Author: pressam chara <pressamchara@gmail.com>
Date:   Mon Oct 19 19:46:11 2026 +0700

    Revert "Welcome to our project"

    This reverts commit 6891530d3eff234b542717b79884ee5dec80e31b, reversing
    changes made to 3b1a67cc65f6da1d3795ea65aae6ac2341490987.

commit 6891530d3eff234b542717b79884ee5dec80e31b
Merge: 3b1a67c 8a4ffb5
Author: pressam chara <pressamchara@gmail.com>
Date:   Mon Oct 19 19:40:59 2026 +0700

    Welcome to our project

commit 8a4ffb59d849ab6886dbd5e870024252fd176aa6 (feature/greeting)
Author: pressam chara <pressamchara@gmail.com>
Date:   Mon Oct 19 19:36:46 2026 +0700

    Add personalized greeting feature

commit 3b1a67cc65f6da1d3795ea65aae6ac2341490987
Author: pressam chara <pressamchara@gmail.com>
Date:   Mon Oct 19 19:35:19 2026 +0700

    Initial commit: GitHello programming

commit e4ed821ddef1a2f3fd7a215a3e5e34e4d5f23544
Author: pressam chara <pressamchara@gmail.com>
Date:   Mon Oct 19 19:21:41 2026 +0700


User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ git log --oneline
f789b78 (HEAD -> master) Revert "Welcome to our project"
6891530 Welcome to our project
8a4ffb5 (feature/greeting) Add personalized greeting feature
3b1a67c Initial commit: GitHello programming
e4ed821 Welcome to our project
1a6dcde Add personalized greeting feature
59d9b9e Add personalized greeting feature
1b5c1f7 Initial commit: GitHello programing

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ ls
GitHello  GitHello.java  HelloGit.java  yy
```

Practice 5. Use .gitignore for Java Builds

```
User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ ls
GitHello  GitHello.java  HelloGit.java  yy

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ cat GitHello.java
public class HelloGit { public static void main(String[] args) { System.out.println("Hello, Git!"); } } System.out.println("Hello, " + args[0] + "!"); Syste
m.out.println("Hello, " + args[0] + "!"); public class HelloGit { public static void main(String[] args) { System.out.println("Hello, Git!"); } }

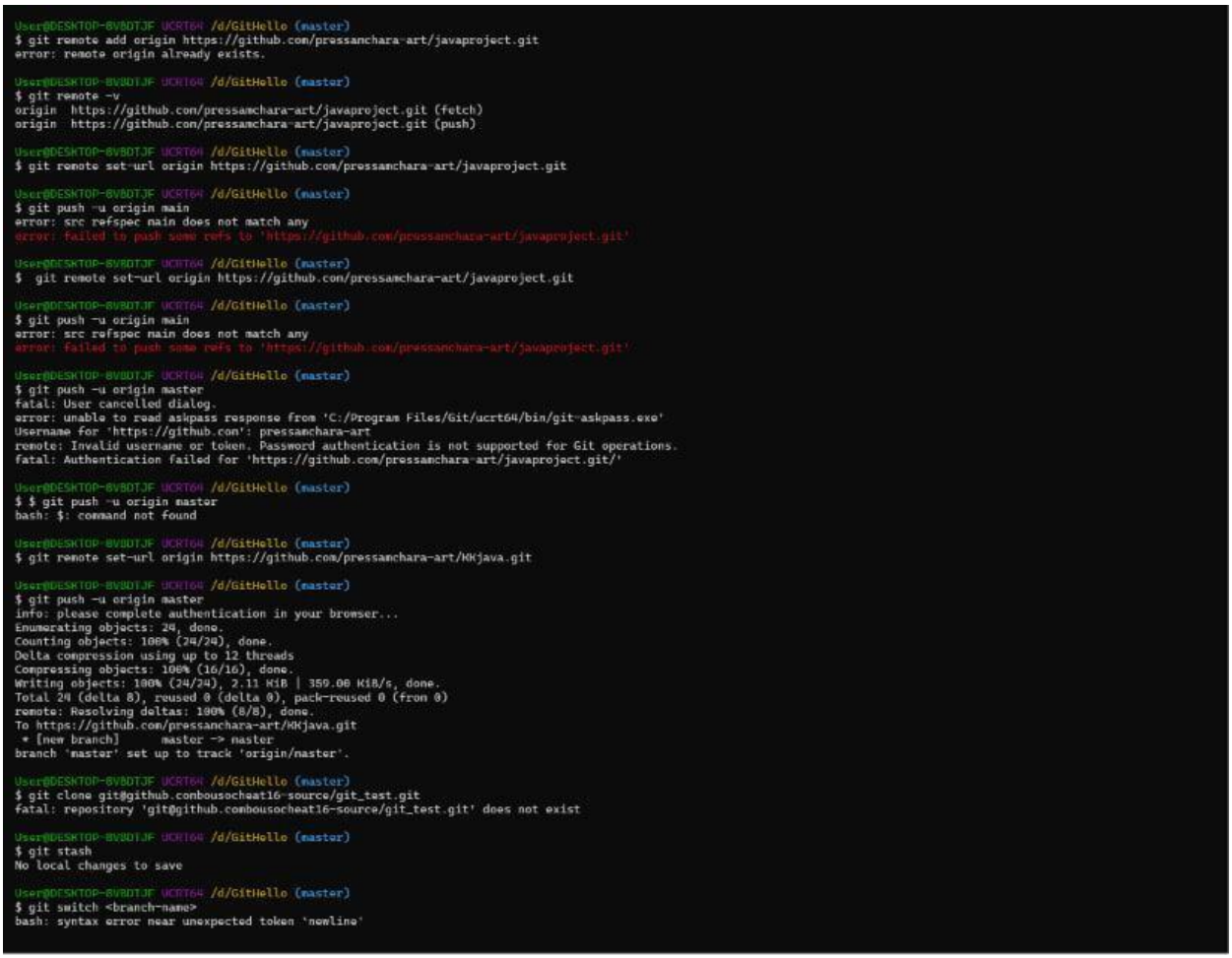
User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ nano  .gitignore

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ git rm -r --cached .
rm 'GitHello.java'

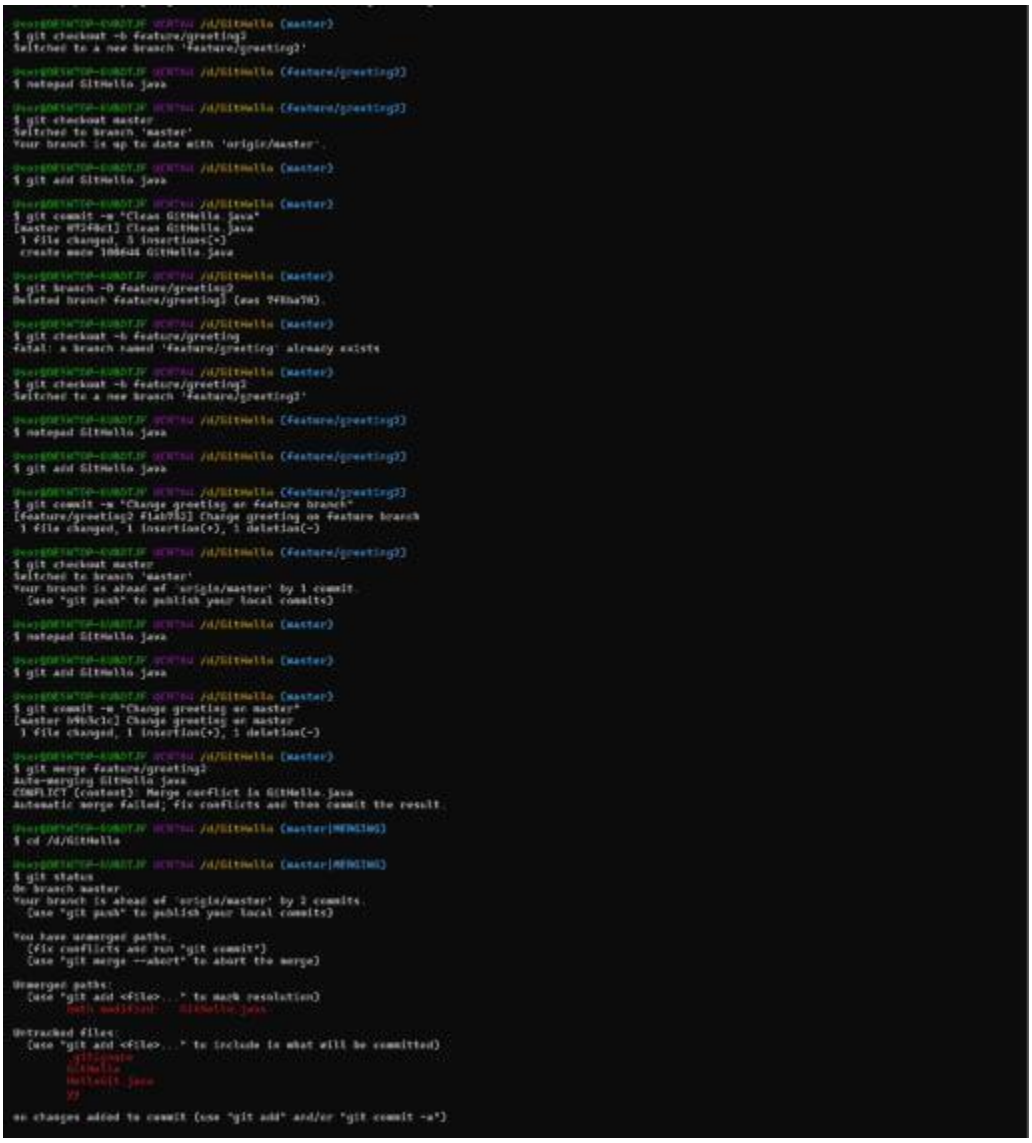
User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ git add
Nothing specified, nothing added.
hint: Maybe you wanted to say 'git add .'?
hint: Disable this message with "git config set advice.addEmptyPathspec false"

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ git commit -m "Added .gitignore file"
[master 7f8ba70] Added .gitignore file
 1 file changed, 1 deletion(-)
 delete mode 100644 GitHello.java
```

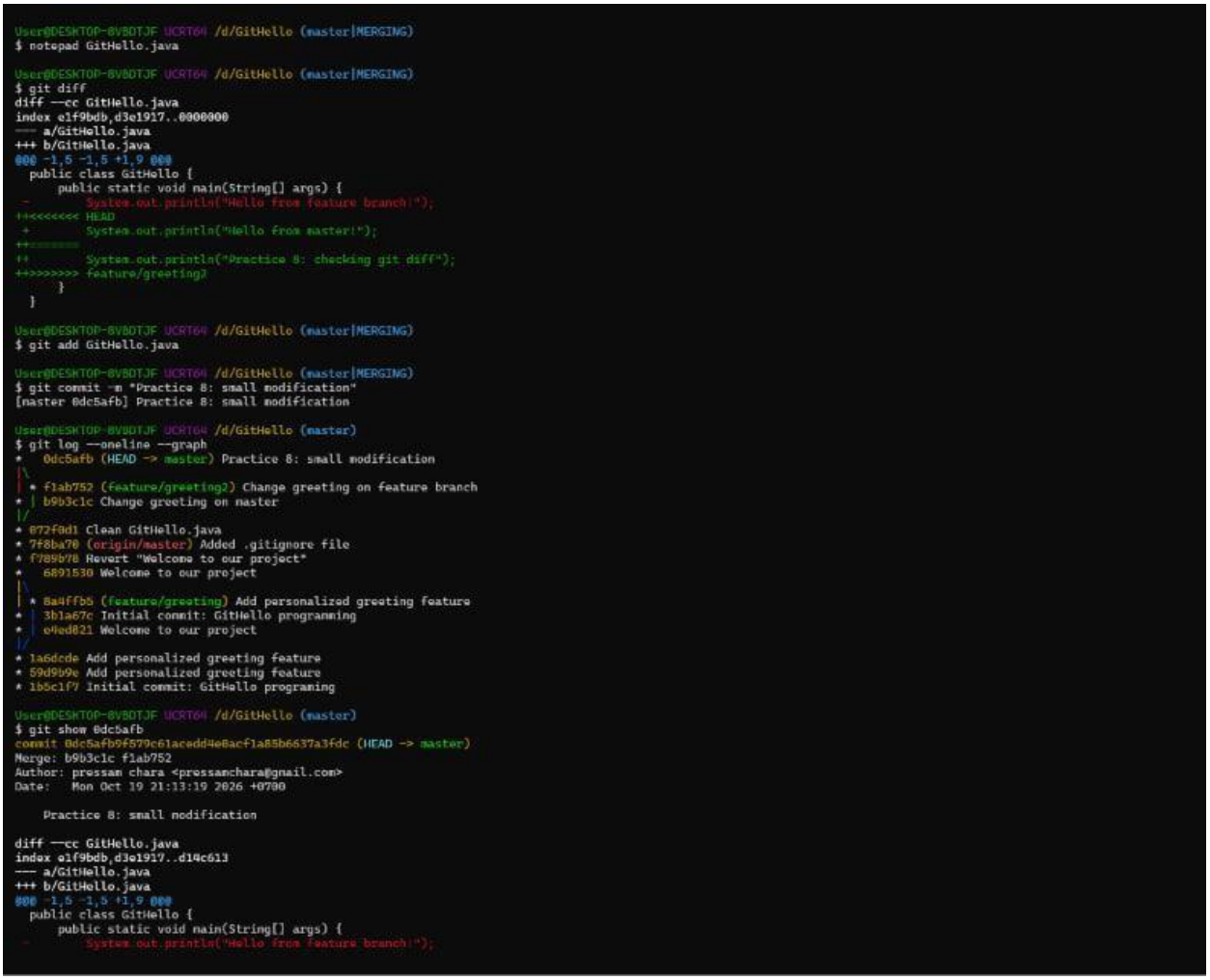
Practice 6. Collaborate via Remote Repository on the cloud



Practice 7. Resolve a Merge Conflict



Practice 8. Review Changes with git diff and git log



```
User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master|MERGING)
$ notepad GitHello.java

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master|MERGING)
$ git diff
diff --cc GitHello.java
index e1f9bdb,d3e1917..0000000
--- a/GitHello.java
+++ b/GitHello.java
@@@ -1,5 -1,5 +1,9 @@@
  public class GitHello {
      public static void main(String[] args) {
-         System.out.println("Hello from feature branch!");
++<<<<<<< HEAD
+         System.out.println("Hello from master!");
++=======
++        System.out.println("Practice 8: checking git diff");
++>>>>>>> feature/greeting2
      }
  }

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master|MERGING)
$ git add GitHello.java

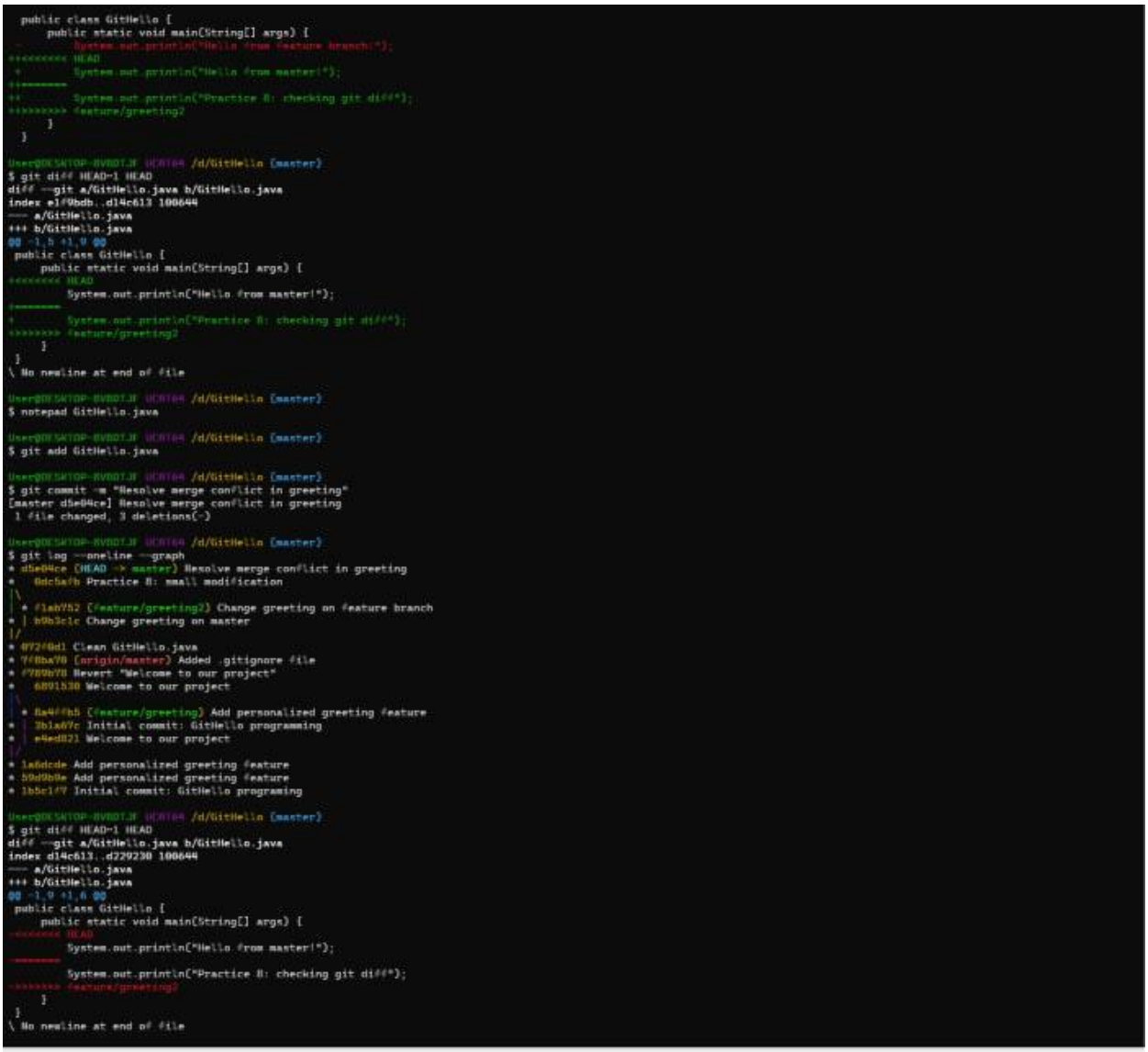
User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master|MERGING)
$ git commit -m "Practice 8: small modification"
[master 0dc5afb] Practice 8: small modification

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ git log --oneline --graph
*   0dc5afb (HEAD -> master) Practice 8: small modification
|\
| * f1ab752 (feature/greeting2) Change greeting on feature branch
* | b9b3c1c Change greeting on master
|/
* 872f0d1 Clean GitHello.java
* 7f8ba70 (origin/master) Added .gitignore file
* f789b78 Revert "Welcome to our project"
*   6891530 Welcome to our project
|\
| * 8a4ffb8 (feature/greeting) Add personalized greeting feature
* | 3b1a67c Initial commit: GitHello programming
* | e4ed021 Welcome to our project
|/
* 1a6dcde Add personalized greeting feature
* 59d9b9e Add personalized greeting feature
* 1b5c1f7 Initial commit: GitHello programing

User@DESKTOP-8VBDTJF UCRT64 /d/GitHello (master)
$ git show 0dc5afb
commit 0dc5afb9f579c61acedd4e8acf1a85b6637a3fdc (HEAD -> master)
Merge: b9b3c1c f1ab752
Author: prossam chara <prossamchara@gmail.com>
Date:   Mon Oct 19 21:13:19 2026 +0700

    Practice 8: small modification

diff --cc GitHello.java
index e1f9bdb,d3e1917..d14c613
--- a/GitHello.java
+++ b/GitHello.java
@@@ -1,5 -1,5 +1,9 @@@
  public class GitHello {
      public static void main(String[] args) {
-         System.out.println("Hello from feature branch!");
```



Practice 9: Let redo your Lab1 exericse. Use git to track your code change. When you are done exercise 1, add and commit it to your local repo. Repeat for all other exercises. Finally, create a repo on gitlab/github. Push all this codes in your current work to this git repo to store it on the remote repository in the cloud.